

Advanced BGP: Anycast, Policy, and Internet Presence

1 IP-Anycast

BGP is not just for standard routing; it also implements **IP-anycast**, which is crucial for services like DNS.

IP-Anycast Mechanism

- **Goal:** Replicate the same content on geographically dispersed servers and route each user to the "nearest" one.
- **Configuration:** Every server is assigned the **same IP address**.
- **BGP Role:** Each server advertises this IP. BGP routers receive multiple advertisements for the same address, treating them as different paths to the same location.
- **Selection:** The BGP route-selection algorithm naturally picks the "best" (closest) server based on AS hops or other metrics.

Primary Use Case: DNS

While CDNs often avoid anycast due to TCP session instability (packets of the same connection might hitting different servers), the **DNS system** uses it extensively. DNS root servers (13 IP addresses) are actually hundreds of physical servers worldwide, reached via anycast.

2 Routing Policy

In inter-AS routing, **Commercial Relationships** often trump shortest paths. Policies are enforced via selective BGP advertisements.

Common BGP Policies

- **Stub Networks (Access ISPs):** They advertise prefix reachability only for themselves to avoid carrying transit traffic between backbone providers.
- **Backbone Rule of Thumb:** An ISP usually only carries transit traffic if the source or destination (or both) is one of its **paying customers**.

- **Peering Agreements:** Backbone providers (A, B, C) might carry each other's traffic but not act as a free "middle-man" for a third backbone provider.

3 Inter-AS vs. Intra-AS: The Differences

Feature	Inter-AS (BGP)	Intra-AS (OSPF/RIP)
Priority	Policy is dominant. Decisions are commercial and administrative.	Performance is dominant. Focused on shortest/cheapest path.
Scale	Must handle hundreds of millions of routers; uses AS clusters.	Smaller scale; hierarchy built by areas.
Cost	No notion of cost (other than hops/policy).	Explicit link weights/costs.

Table 1: Comparison of Routing Protocol Philosophies

4 The Big Picture: Obtaining Internet Presence

Establishing a presence on the Internet involves a sequence of protocols working in harmony:

End-to-End Presence Workflow

1. **Connectivity:** Contract with a local ISP → Physical connection → Assigned an IP prefix (e.g., /24).
2. **Names:** Contract with a Registrar (e.g., ICANN-accredited) → Obtain Domain (Xanadu.com).
3. **DNS Presence:** Register your DNS server's IP with the Registrar → Entry added to TLD servers (.com).
4. **Internal Config:** Map hostnames (www.xanadu.com) to IPs in your local DNS server.
5. **Global Reach (BGP):** Your local ISP uses **BGP** to advertise your /24 prefix to its neighbors. The advertisement propagates globally, informing all routers how to reach your network.